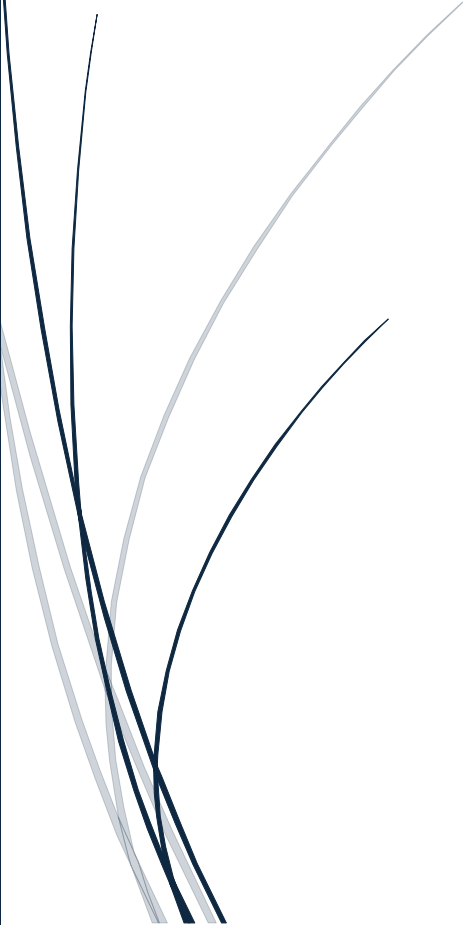




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Portfolio Deep Learning

Max ter Steege



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Portfolio assignment 1 – Max ter Steege

Evaluating ethical models & application on data ethics

Various ethical approaches provide a framework to resolve moral dilemmas, such as a 'Deontology' approach (Misselbrook, 2013). The deontology approach is mainly influenced by Immanuel Kant and evaluates actions based on how they apply to universal actions, duties and principles, independent of the outcome. In other words, an action is ethical if it is morally correct. Virtue ethics offers a different approach. Mainly influenced by Aristotle, the focus lies on character and being a virtuous person (Van Hooft, 2014). The main thought behind being a virtuous person according to Aristotle, is to achieve 'Eudaimonia' which refers to the ultimate goal; to achieve human flourishing (Huta, 2013). An action is evaluated based on topics like integrity, responsibility, fairness and how it contributes to achieving eudaimonia. Therefore, an action is ethical if it is virtuous and contributes to achieving the ultimate goal.

On a finer granularity, ethical models provide a framework on how to handle moral cases, such as Impact Assessment Mensenrechten en Algoritmes (IAMA) and Koninklijke Nederlandsche Maatschappij tot bevordering der Geneeskunst (KNMG).

The IAMA framework (Ministerie van Binnenlandse Zaken en Koninkrijksrelaties, 2021) focuses on assessing the human rights impact of algorithmic systems before and during deployment. It evaluates aspects such as discrimination, transparency, and potential societal harms. The purpose of IAMA is to identify risks as early as possible and ensure that the developed algorithmic systems respect fundamental rights. An action is considered ethical when the algorithmic system can be justified in terms of human rights, transparency, and responsible governance.

The KNMG (KNMG, n.d.) ethical framework approaches moral dilemmas from a medical decision making perspective. The model outlines six clear steps on how to handle moral cases. The emphasis lies on careful moral reflection rather than on a purely technical risk assessment. The KNMG approach encourages users to examine a case from multiple perspectives, weigh conflicting values, and reach a justified ethical decision.

Both models provide a framework on how to handle a moral dilemma, but differ in their application and in their field of practice. The IAMA framework is a bit more bureaucratic (as it finds its origin in the government), while the KNMG model can be seen as a more practical approach (as it finds its origin in the medical field).

Suppose a hospital considers implementing an algorithm that predicts the remaining life expectancy of patients. Using the IAMA framework, the primary focus would be on evaluating risks such as bias in the training data, transparency of the model, possible discrimination, and the impact on fundamental rights. The goal would be to determine

whether the algorithm can be deployed responsibly and which safeguards are required (all in all quite bureaucratic without a real consideration for ethical implications).

Using the KNMG framework, the focus would lie more on the ethical implications, such as whether communicating model predictions to patients would be worth it (which is a practice that follows the main thought of virtue ethics). Medical professionals would have a discussion about questions such as whether providing this information benefits the patient, whether it could cause harm, and how it affects patient dignity.

The outcome of both approaches could lead to similar conclusions, but the reasoning behind the decision differs. In my view, while any situation is highly context dependent, the KNMG ethical approach is generally more useful for data ethics dilemmas. Ethical situations are often highly context dependent, and rule based approaches do not always capture the complexity of real world situations. This does not mean that frameworks such as IAMA (in a sense a deontology approach) are useless, but that the application of these frameworks is context dependent as well. For example, IAMA could be more useful for identifying specific detailed risks in vast algorithmic systems. Ultimately, ethical decision making more often benefits from combining a structured framework with context dependent moral issues in data ethics cases and therefore I prefer the KNMG approach.

Portfolio assignment 2 – Max ter Steege

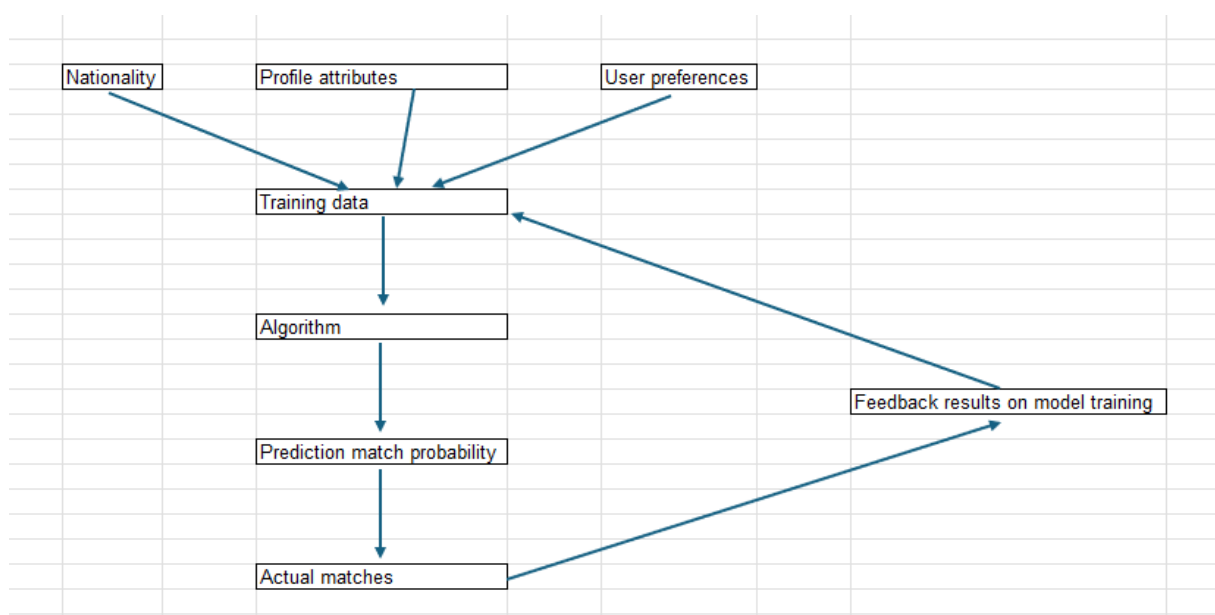
A Dutch dating app needs to adjust their algorithm to prevent discrimination. It was decided that the developers need to adjust their algorithm to make sure that non-Dutch users are being shown or matched as much as Dutch users (College voor de Rechten van de Mens, 2023).

My first impression is that the issue likely comes from the training data. Dating algorithms usually learn from past user behavior. If users historically liked or matched more often with certain groups, the model will learn that pattern. Lower visibility for non-Dutch users does not automatically mean the algorithm was designed to discriminate. It probably just means that the training data contained fewer samples of non-Dutch matches. Also, fewer matches with non-Dutch users can also be linked to user preferences of Dutch users such as profile quality, hobbies, language, or attractiveness. At the same time, there is a difference between individual preferences and fairness on a group level. For example, filtering by gender in a heterosexual dating app is functional, not discriminatory.

I doubt that forcing equal visibility is the best solution. It changes the results of the model instead of improving the model itself, and this may lower its performance. A better approach would be to first check whether certain groups are actually under represented for unfair reasons. This can be done by analyzing the data and letting humans review the results (semi supervised learning). If a real bias is found, specific adjustments can be made, instead of forcing full 'equality' for everyone (which may lower actual match percentages).

Figure 1

DAG – Dilemma algorithm dating app



I don't see any direct difference with the first impression after drawing the DAG.

If a data scientist gets an assignment like this, the first step should be to understand how the system actually works before changing anything. There needs to be a check whether the problem is caused by the model itself or by actual user behavior/preferences. Then measure whether there is real unfairness in visibility or matching rates. If action is needed, choose a clear fairness goal and apply targeted adjustments instead of forcing equal results without analysis. Finally, document the tradeoff between fairness and model performance.

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